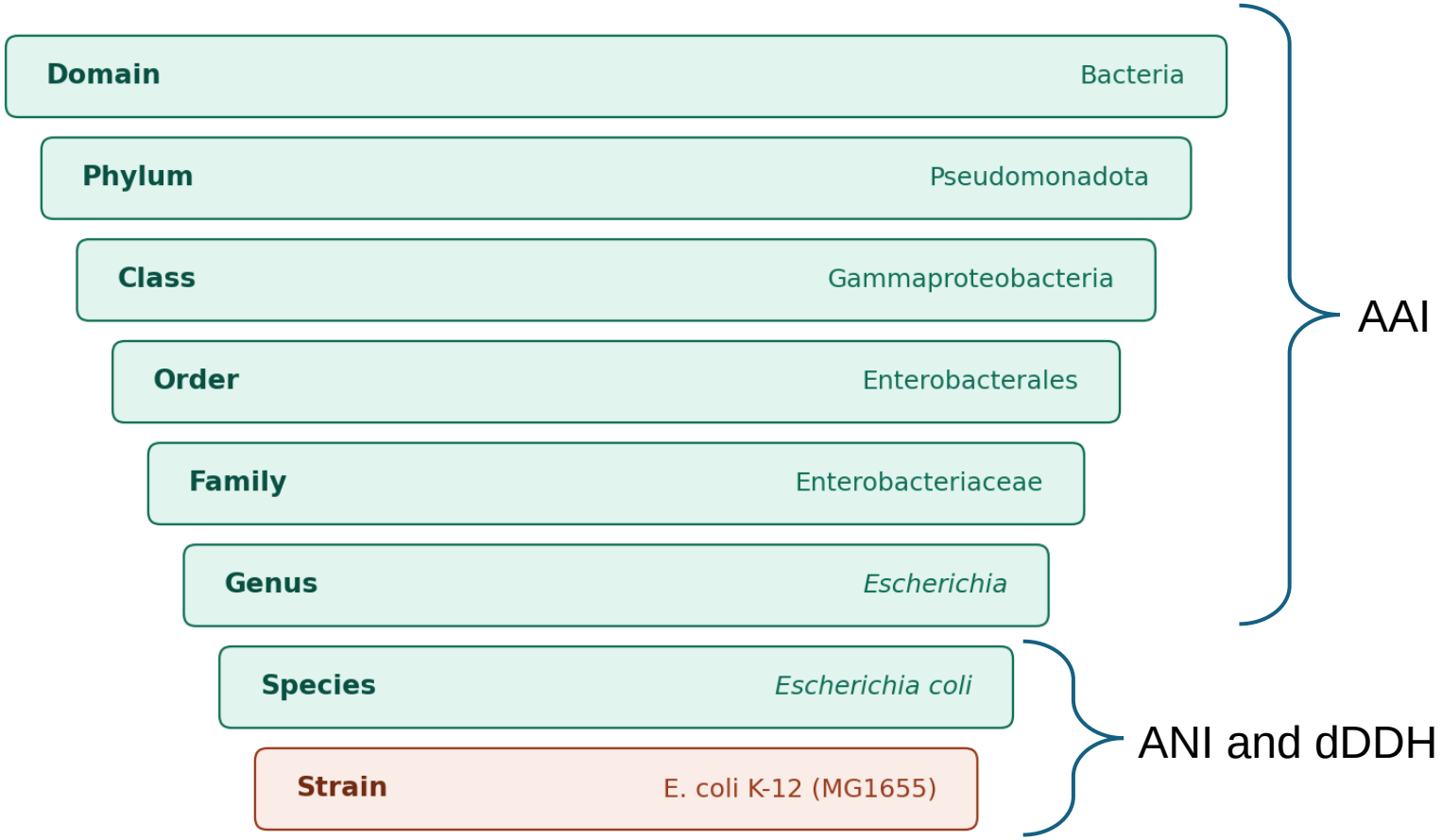
Day 6: Comparative genomics

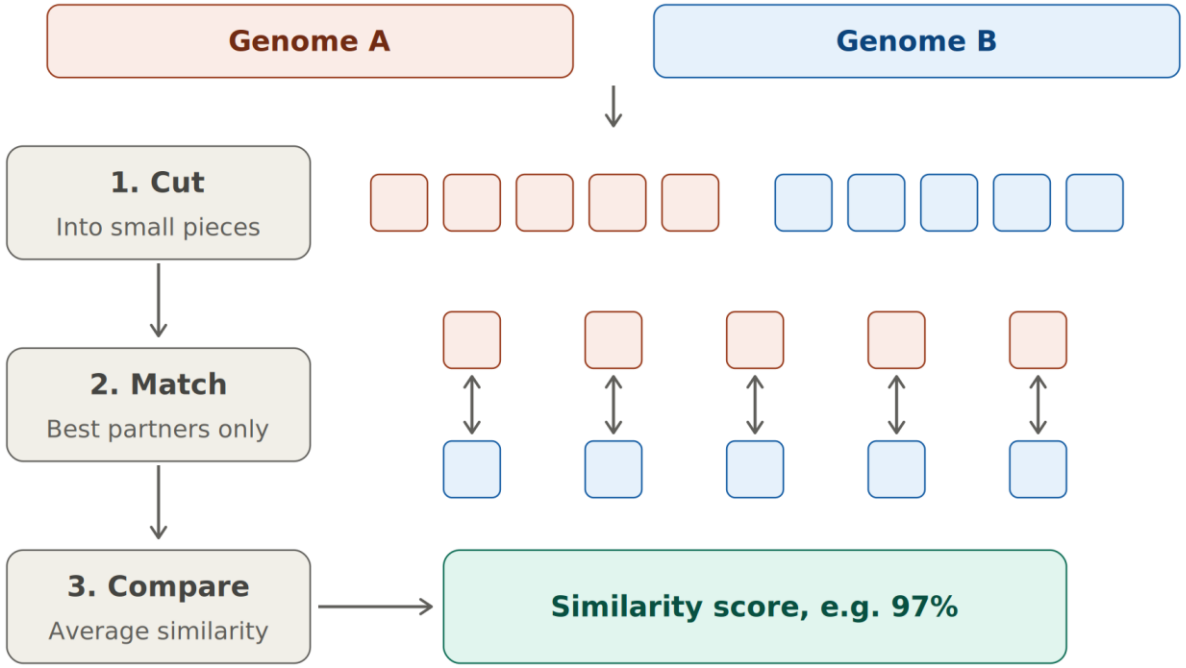
MMB-127

Genome related indexes to classify and compare bacteria



Genome related indexes to classify and compare bacteria

- Average nucleotide identity (ANI)

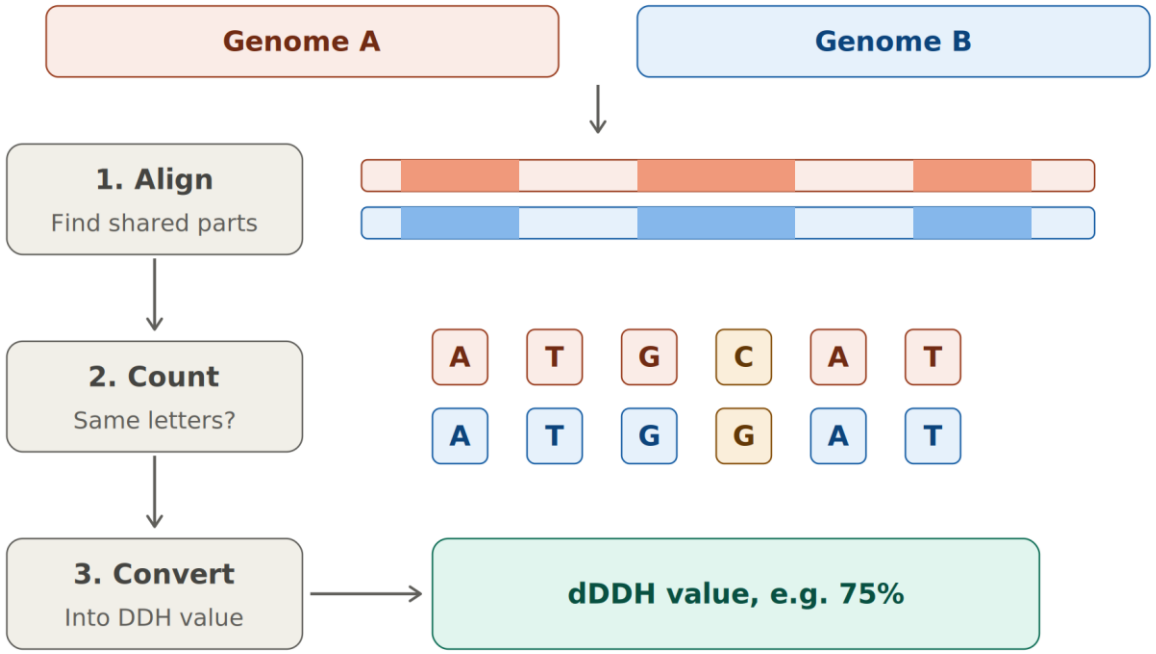


Taxonomic rank	ANI/AAI threshold
(Same) Strain	ANI > 99.99%, gene-content >99% (typical)
Genomovar or sequence type	ANI > 99.5%, gene-content > 90% (typical)
Species	ANI > 95.0%, gene-content > 70% (typical)
Genus	AAI > 65.0%, gene-content > 50% (typical)
Family	AAI > 45.0%, gene-content > 20% (typical)

Konstantinidis, 2023

Genome related indexes to classify and compare bacteria

- Digital DNA-DNA hybridization



- Formula 1
- Formula 2 = the only one robust to incomplete genomes
- Formula 3 = Complete/closed genomes

>70% = same species

Auch et al., 2010